Ray Sarabu

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Education

Carnegie Mellon University — Pittsburgh, PA
B.S. Electrical and Computer Engineering + Robotics; GPA: 3.65/4.0 — *Aug 2024 – May 2027*

- **Relevant coursework:** Robotics Systems Engineering, Spacecraft Design-Build-Fly, Machine Learning for Engineers, General Robotics, Feedback Control Systems, Digital System Design, Embedded Systems, Computer Graphics
- **Teaching Assistant,** Reasoning with Data (36-200): led weekly labs and office hours for 40+ students covering statistical reasoning, R programming, and data analysis

Professional Experience

General Motors — Autonomous Vehicle Systems Integration Team — Warren, MI
Advanced Systems Development Intern — *May – Aug 2026*

- Helped build GM's ADAS vehicle-checkout tool for automated setup and verification, implementing 35 workflow steps and 76 pass/fail checks across 5 LiDARs, 8 radars, and 11 cameras to replace a manual pre-run checklist.
- Built a visualization tool that parses 128-channel LiDAR UDP multicast streams in real time and reconstructs full 3D point clouds directly from raw MSOP sensor packets.
- Applied camera–LiDAR sensor fusion to project point clouds onto camera fields of view, giving vehicle operators immediate visual confirmation of sensor alignment and data quality before collection runs.
- Designed and deployed an internal React/Node operations dashboard that polls live build data every 30s, giving build teams real-time visibility into blockers, per-vehicle build status, and issue ownership.

CMU MetaMobility Lab — Wearable Robotic Knee Exoskeleton — Pittsburgh, PA
Student Researcher — *May 2025 – Apr 2026*

- Prototyped 4 functional generations of the lab's first robotic knee exoskeleton, improving embedded electronics, mechanical integration, and hardware design to achieve 20 N·m of torque transfer with better wearability.
- Engineered the embedded control stack on Teensy and ESP32 microcontrollers, integrating IMU sensing and SPI/I2C/UART/CAN communication for real-time, low-latency robotic control.
- Implemented over-the-air (OTA) firmware updates, enabling rapid controller iteration during human-subject testing without disassembling the device.
- Developed impedance-based controllers and spline trajectory generation to align exoskeleton torque assistance with the natural walking gait cycle.

Projects

Arm-adillo Rolling Manipulator | *MuJoCo, Reinforcement Learning* — Sep 2026 – Present

- Designed a 6-servo robot arm that needs no mobile base: its curved links curl into a wheel so it can roll itself across the floor, then unfold to pick up objects and carry them inside the ring, reaching 53 cm and carrying 300 g payloads in simulation.
- Trained residual PPO policies (Stable-Baselines3, 128×128 MLP) that correct a scripted rolling gait under per-episode domain randomization, rolling 2.7× farther on 40% less servo heat per meter across 16 randomized robots.
- Building the physical prototype from a 150-variant optimized link geometry: 3D-printed curved link shells and servo brackets, assembled the servo chain, and began measuring the hardware to recalibrate the simulation model.

Argus 3 Nanosatellite | *Vision-Based GNC, Embedded ML* — Aug 2026 – Present

- Deployed 27M- and 105M-parameter Earth-imagery localization models to the CubeSat's Jetson Orin Nano flight computer through a PyTorch–ONNX–TensorRT pipeline, reaching 15/44 ms per-image inference validated against real ISS photographs.
- Developing and testing an onboard imaging agent that schedules the next photo capture from predicted ground track, sun angle, eclipse state, and expected image quality, validating it in simulation before flight deployment.
- Building a CubeSat digital twin for software/hardware-in-the-loop testing, pairing a headless Basilisk dynamics core with Unity rendering of NASA imagery; built its SPICE layer for Sun/Earth ephemerides and eclipse timing.

Self-Balancing Robot | *IMU Odometry, PID Control* — Jan – May 2026

- Designed and tuned a self-balancing differential-drive robot using IMU odometry and PID control to reject push disturbances.
- Fused accelerometer and gyroscope data with a two-state Kalman filter for drift-corrected tilt estimation at 100 Hz.

Search-and-Rescue Robot | *Computer Vision, Autonomous Navigation* — Jan – May 2026

- Designed and built a rugged urban search-and-rescue (USAR) robot for disaster-response environments, scaling stairs, climbing 45° inclines, and traversing debris-strewn terrain while surviving 3 ft impact drops.
- Built an OpenCV vision, line-following, and odometry localization stack for autonomous navigation to simulated victims.

Skills

Programming: Python, C, C++, Java, JavaScript, R, SystemVerilog

Embedded & Robotics: ESP32, Teensy, Raspberry Pi, SPI/I2C/UART/CAN, PCB design, IMU sensing, PID/impedance control

Tools & Frameworks: OpenCV, NumPy, PyTorch, MuJoCo, React, Node.js, Linux, Git, KiCad, SolidWorks, Fusion 360, Inventor

Activities: Choreography Chair (Greek Sing), CMU Men's Volleyball, Phi Delta Theta, weightlifting, cooking, pickleball